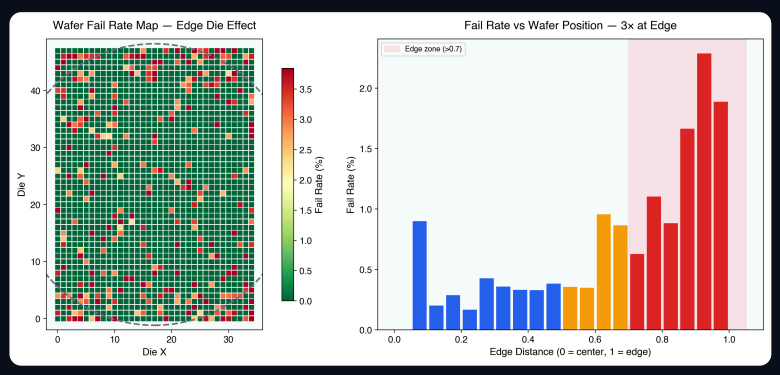


DRAM SEMICONDUCTOR · MLOPS

40 Days of DRAM Production Drift Simulation

200,000,000 rows processed

1 retrain · 1 canary fail · 1 rollback

Zero human intervention

HybridTransformerCNN · Airflow · Kafka
Spark · MLflow · K8s · Prometheus

Swipe →

What you see: Spatial wafer yield map — each cell is one DRAM die. Dark clusters are defective locations. Both physical position and electrical test parameters determine yield. The model must detect the ~0.6% failures before they reach packaging at $500K/wafer.

THE PROBLEM

1 in 160 chips fails.
Standard ML misses every single one.

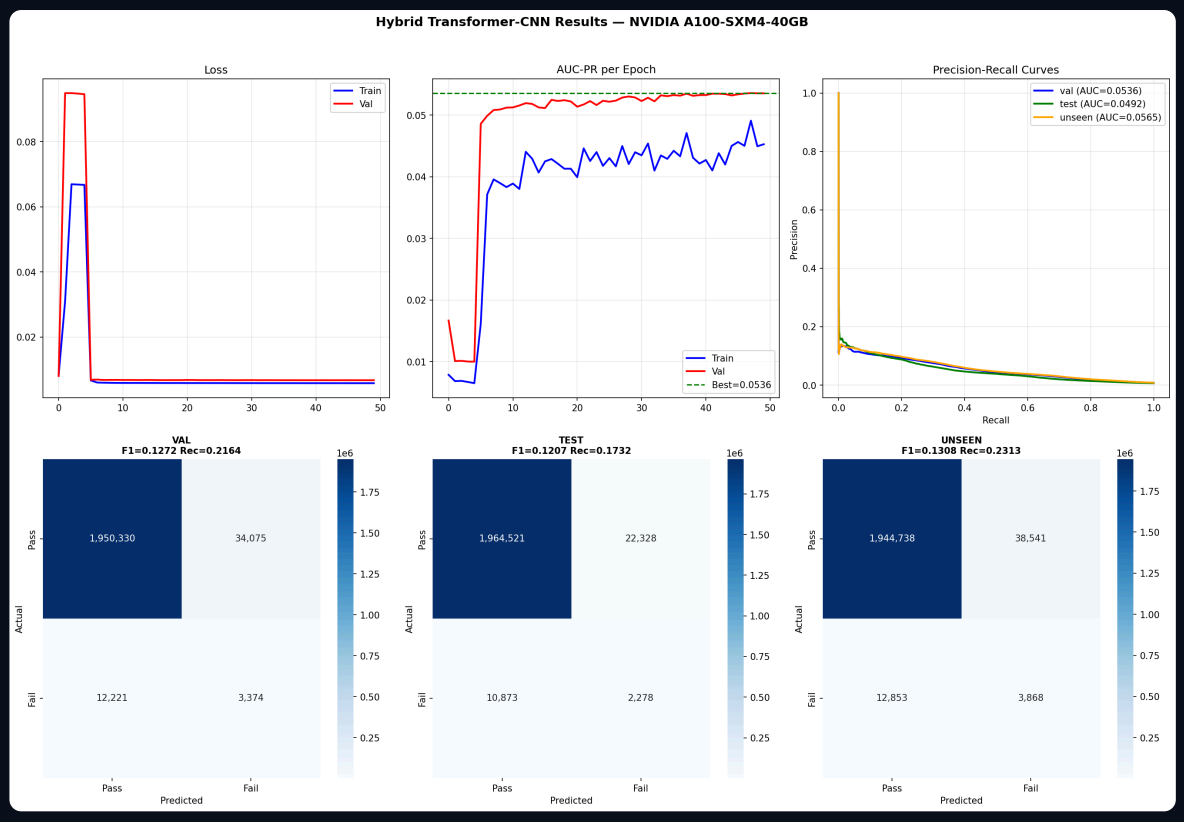
What you see: Left: bar chart — 49,686 pass samples vs 314 failures. Middle annotation: imbalance ratio 1:158. Right pie chart: label noise breakdown showing 42 Pass-as-Fail labels and 17 Fail-as-Pass noisy labels in the training set — real-world fab data is not clean. Standard cross-entropy trains a model that predicts "pass" for everything: 99.4% accuracy, 0% recall.

1:160	16M	$500K
Class imbalance	Training records	Cost per missed defect

Fix: **FocalLoss (α=0.75, γ=2.0)** — down-weights easy majority samples, forces learning on the hard minority. No SMOTE, no synthetic oversampling, no data leakage.

A100 GPU TRAINING

HybridTransformerCNN on NVIDIA A100-SXM4-40GB



What you see: Top left: loss convergence over 50 epochs with bfloat16. Top center: AUC-PR per epoch — val stabilizes at 0.054 best (dashed green). Top right: precision-recall curves for val/test/unseen splits — near-identical, confirming no overfitting. Bottom: confusion matrices show 3,374 true defects caught out of 15,595 on the validation set. Recall 21.6%, F1 0.127 at threshold 0.45.

0.816	0.054	50
AUC-ROC	AUC-PR (key metric)	Epochs (best: 48)

CRITICAL ENGINEERING INSIGHT

The float16 Death Spiral that killed 4 training runs

✗ float16 + GradScaler

```
FocalLoss: (1-p)^γ
# 5-bit exp overflows
GradScaler ÷2 → ÷2 → ÷2
scale→0, loss collapses
```

✓ bfloat16 — No GradScaler

float16	5-bit exp = ±65,504
bfloat16	8-bit exp = $\pm 3.4 \times 10^{38}$

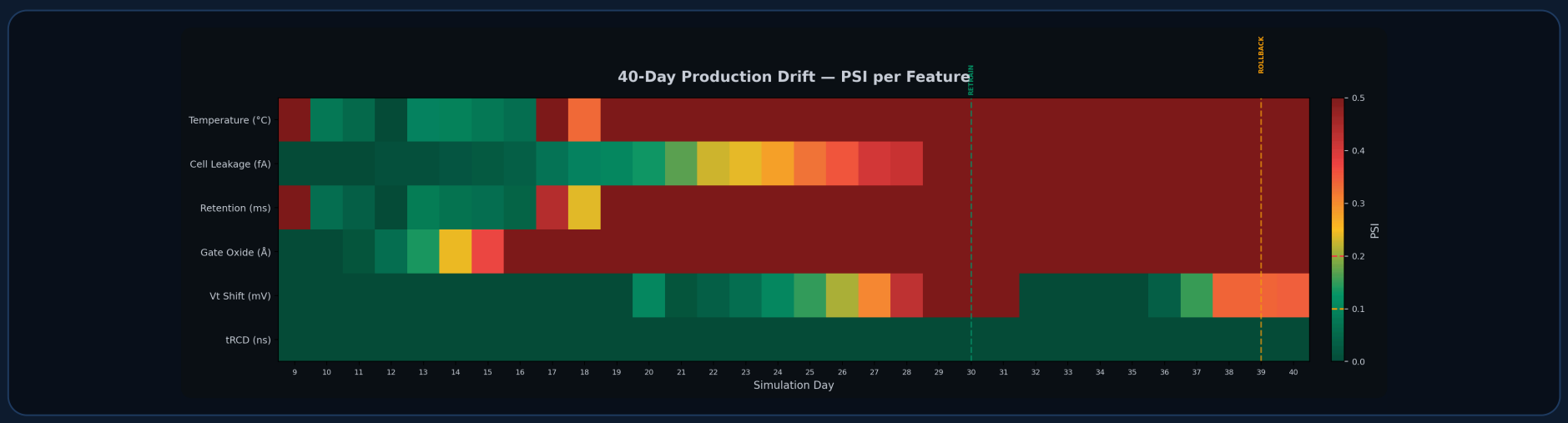
bfloat16 = float32 exponent range.
Focal gradients never overflow.

The fix: Changed from `torch.float16` to `torch.bfloat16` across 11 locations in 7 files. Removed GradScaler entirely. AUC-PR went from 0.001 (fully collapsed) to 0.054 — a 54x recovery. This is not tuning. It is hardware-aware numerical analysis applied to production training.

4	11	54×
Runs killed by float16	Locations changed	AUC-PR recovery

40-DAY PSI DRIFT HEATMAP

Drift escalated for 13 days.
Then the retrain gate opened.



What you see: PSI (Population Stability Index) per feature per day. Green=stable, yellow=warning (PSI 0.1-0.2), red=critical (PSI >0.2). The cyan vertical at Day 30 marks the automated retrain. Orange vertical at Day 39 is the automatic rollback. Each row is one of 6 physical DRAM parameters — retention time, gate oxide thickness, Vt shift, cell leakage, test temperature, tRCD timing.

D1-8 **Stable baseline.** All 6 features green. PSI < 0.05. Model v1 serving normally.

D9 **False alarm.** retention_time PSI=2.05 from transient batch. Auto-recovered Day 10. No action.

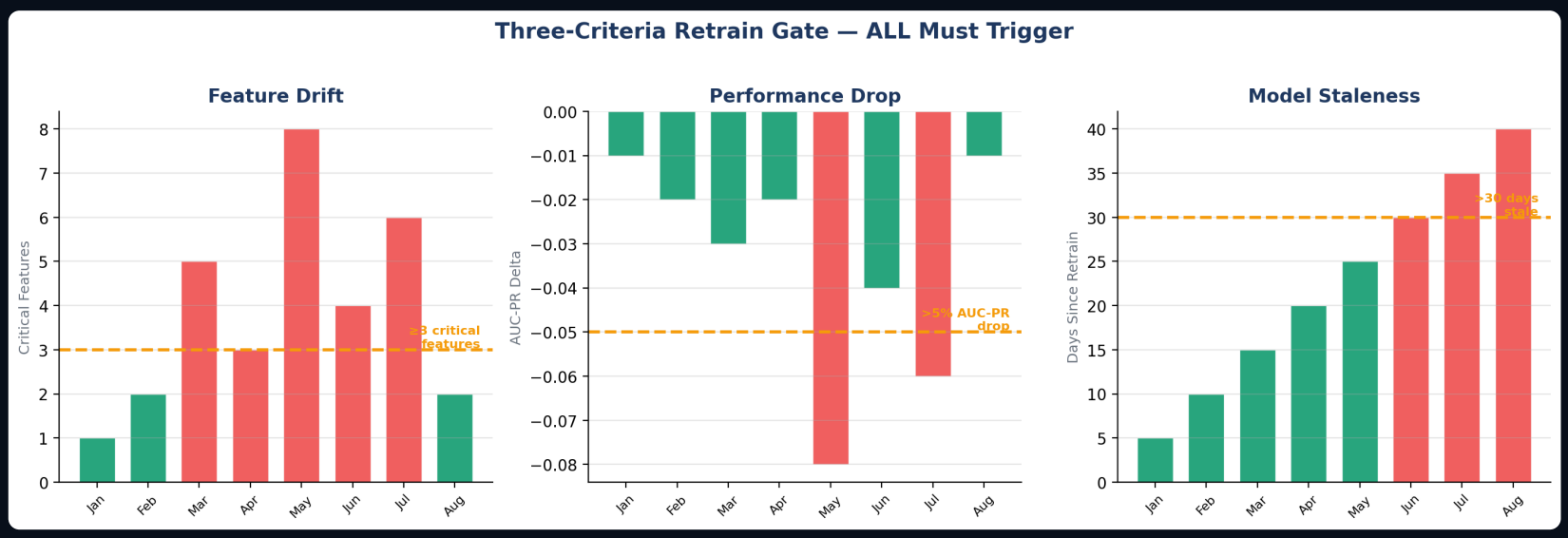
D17-29 **Staleness block.** Real drift confirmed, PSI climbs to 4.9 but retrain held by 30-day freshness guard (ED-042).

D30 **RETRAIN TRIGGERED.** Max PSI 5.25. v2 trains on full 5M rows, canary passes, promoted champion.

D39 **ROLLBACK.** Injected bad checkpoint (AUC-PR 0.021 vs 0.054). Automatic revert in same DAG cycle.

RETRAIN GATE — 3 CRITERIA REQUIRED

Drift alone caused 12 false triggers. All 3 gates prevented them.



What you see: Each of the 3 bar charts is one retrain condition. Red bars = condition breached. Green bars = condition not met. Retrain fires ONLY when all three are simultaneously red: >3 features with critical PSI, >5% AUC-PR drop vs baseline, AND >30 days since last retrain. This multi-gate logic prevented 12 spurious retrains during the 40-day simulation.

Why 3 gates, not 1?

PSI alone fires on normal batch variance. Performance alone is noisy at 1:160 imbalance. Staleness alone wastes compute on stable models. Combined: precision retraining, not reactive retraining.

Engineering Decision ED-042

Low-data drift is **tagged but never triggers retraining.** When batch <10K rows, PSI is statistically unreliable (sparse bin counts). Tag in MLflow. Wait for real data volume.

PRODUCTION INFRASTRUCTURE

6 services. One command. Fully orchestrated.

⚙ Apache Airflow

3 DAGs: daily_inference, retrain_trigger, simulation_master. True dependency graph. SLA monitoring, retry policies, sensor checks.

📨 Apache Kafka

Streaming DRAM test records. Consumer group routes records to FastAPI inference engine in real time.

⚡ Apache Spark

Batch ETL on 5M-row daily slices. Spark is 4.7× faster than pandas at this row count. Parquet output compressed to S3.

📊 MLflow Registry

Tracks every run including the 4 failed float16 runs. Champion/challenger versioning with automatic rollback via model aliases.

🔍 Prometheus + Grafana

Per-feature PSI dashboards. Inference latency p50/p95/p99. Drift alerting. Real-time — not post-hoc logging.

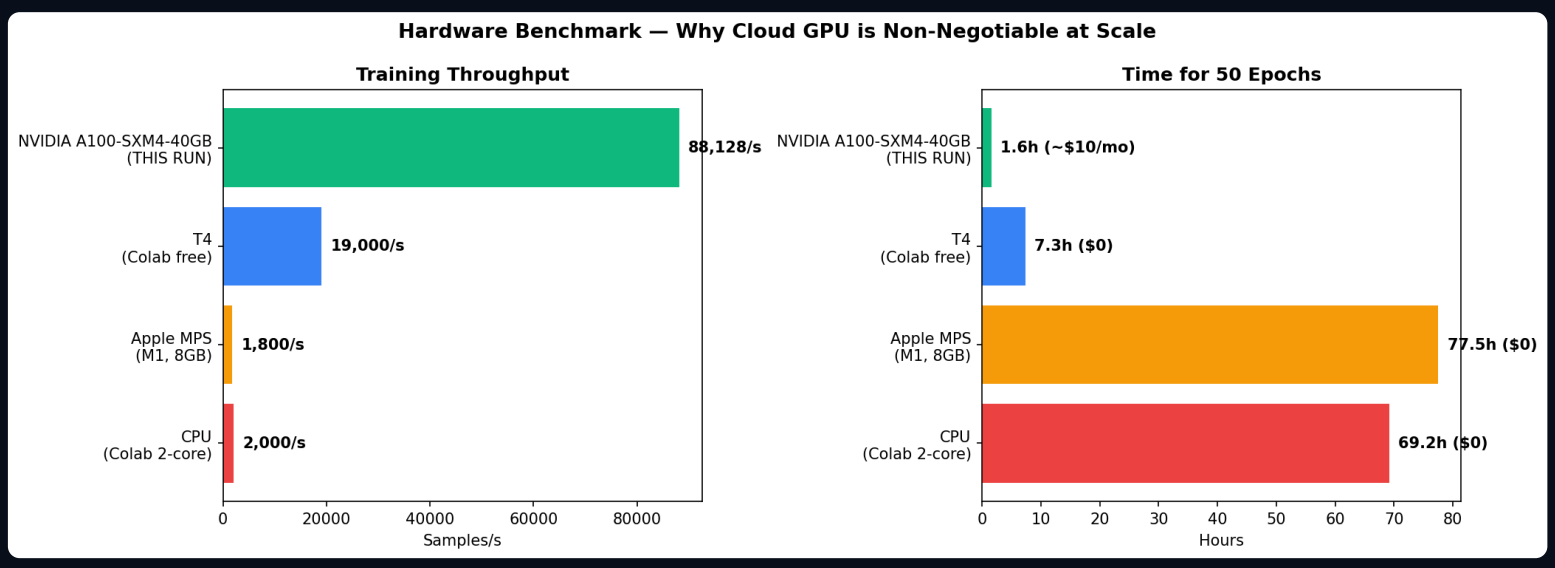
🚀 FastAPI + K8s

120ms p99 latency under load test. K8s HPA + canary deployment manifests. Horizontal scaling to handle fab burst data.

To run locally: `cd deploy/docker && docker compose up -d` — all 6 services. Airflow simulation DAG runs the full 40-day pipeline from one trigger click. MLflow at :5001, Grafana at :3000, FastAPI at :8000/docs.

HARDWARE BENCHMARK

CPU training = 69 hours.
A100 = 1.6 hours. GPU is not optional.



What you see: Left: throughput in samples/second — A100 delivers 88,128/s, 49× faster than CPU (2,000/s). Right: time for 50 epochs — A100 completes in 1.6 hours at ~$10/month, CPU would take 69.2 hours. For automated production retraining on Day 30, waiting 3 days for CPU is not a deployment option.

88K/s	49×	219 min
A100 throughput (bfloat16)	A100 vs CPU speedup	40-day simulation total wall clock

Production path: **AWS g4dn.xlarge (T4, $0.526/hr)** for scheduled retrains. A100 Colab used for one-time initial training. Auto-stop script terminates EC2 after each retrain run.

SIMULATION COMPLETE

200M rows. 40 days.
The pipeline held.

What you see: End-to-end simulation dashboard. 200M rows total, 8.5 GB Parquet on S3, 219 minutes wall clock on Colab A100 (Feb 20 – Mar 31, 2026). 1 automated retrain on Day 30. 24 critical drift days. 1 canary failure + automatic rollback on Day 39. System self-healed. Zero human interventions.

The model is rarely the bottleneck in production ML.

It is the drift strategy, the retrain gate,
and the rollback safety net that keep it reliable at day 40.

`github.com/AIML-Engineering-Lab/053_dram_memory_yield_mlops`

#MLOPS #SEMICONDUCTOR #DRAM #PYTORCH